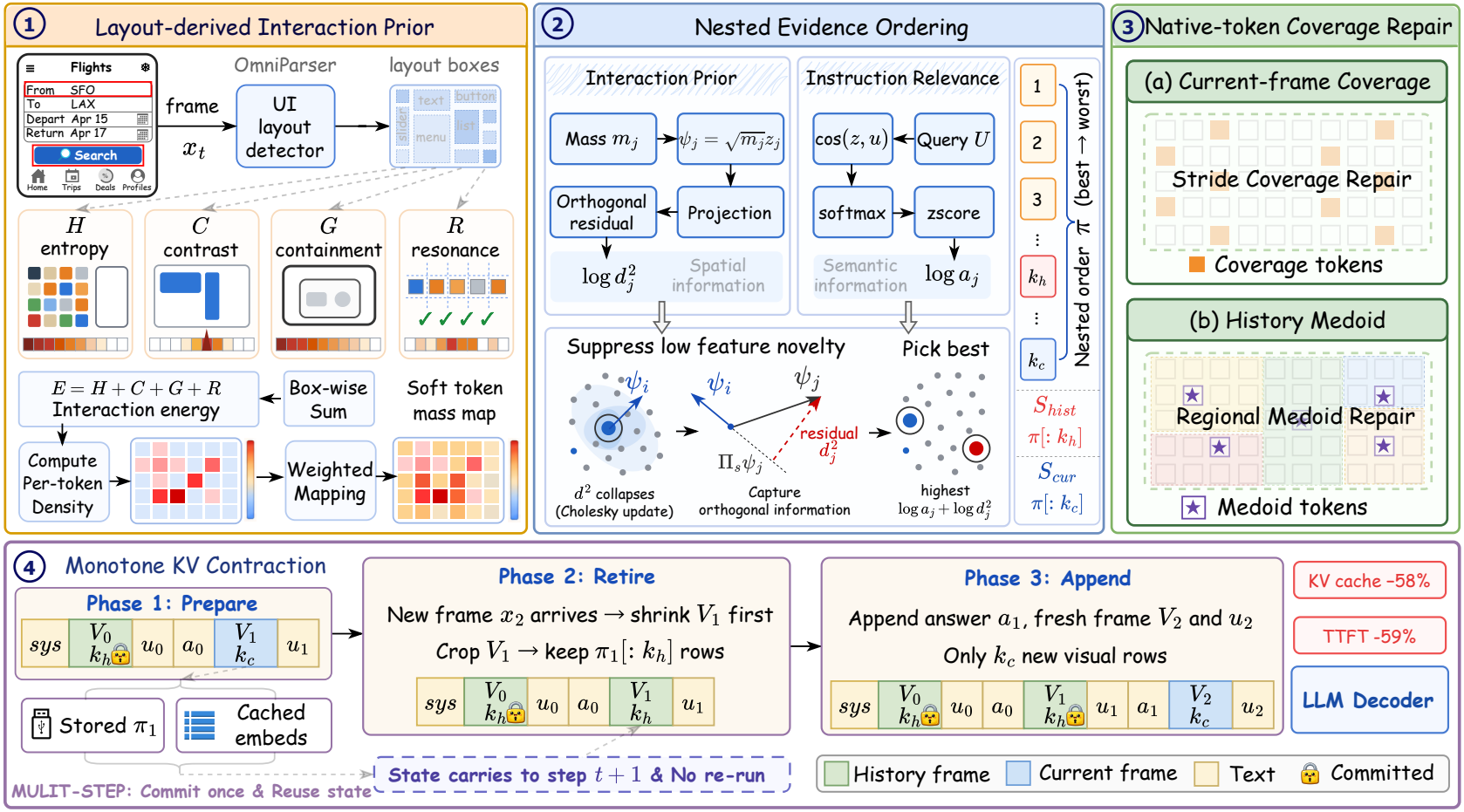

1 Layout-derived Interaction Prior
Flights
From SFO
To LAX
Depart Apr 15
Return Apr 17
Search
Home
Trips
Deals
Profiles
frame
x_t
OmniParser
UI layout detector
layout boxes
text
button
slider
menu
list
H entropy
C contrast
G containment
R resonance
$E = H + C + G + R$
Interaction energy
Box-wise Sum
Soft token mass map
Compute Per-token Density
Weighted Mapping
2 Nested Evidence Ordering
Interaction Prior
Mass m_j
$\psi_j = \sqrt{m_j} z_j$
Orthogonal residual
Projection
$\log d_j^2$
Spatial information
Instruction Relevance
$\cos(z, u)$
Query U
softmax
zscore
Semantic information
$\log a_j$
Suppress low feature novelty
Pick best
ψ_i
ψ_j
residual d_j^2
$\Pi_s \psi_j$
d^2 collapses (Cholesky update)
Capture orthogonal information
highest $\log a_j + \log d_j^2$
1
2
3
k_h
k_c
Nested order π (best → worst)
S_{hist}
$\pi[:k_h]$
S_{cur}
$\pi[:k_c]$
3 Native-token Coverage Repair
(a) Current-frame Coverage
Stride Coverage Repair
Coverage tokens
(b) History Medoid
Regional Medoid Repair
Medoid tokens
4 Monotone KV Contraction
Phase 1: Prepare
sys
V_0 k_h
u_0
a_0
V_1 k_c
u_1
Stored π_1
Cached embeds
MULIT-STEP: Commit once & Reuse state
Phase 2: Retire
New frame x_2 arrives → shrink V_1 first
Crop V_1 → keep $\pi_1[:k_h]$ rows
V_1 k_h
State carries to step $t+1$ & No re-run
Phase 3: Append
Append answer a_1, fresh frame V_2 and u_2
Only k_c new visual rows
a_1
V_2 k_c
u_2
KV cache -58%
TTFT -59%
LLM Decoder
History frame
Current frame
Text
Committed